AEGIS KNOWLEDGE INTELLIGENCE SYSTEM

Provenance-Preserving Ingestion, Conflict Reconciliation & Evidence QA Architecture

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1. Executive Summary

The **Aegis Knowledge Intelligence System** addresses a critical challenge in industrial intelligence: transforming messy, heterogeneous, and partially contradictory technical documentation into a strictly grounded, provenance-preserving knowledge base without hallucination. Built specifically for the fictional Aegis Series-7 Hydraulic Control System (HCS), the architecture processes 20 files spanning 8 distinct formats (born-digital PDFs, scanned image PDFs, raster screenshots, vector schematics, Word docs, Excel registers, PowerPoint slides, and machine JSON dumps). The system achieves **100% benchmark accuracy** across all 23 evaluation questions, **100% citation validity**, and **100% abstention precision** on negative and unanswerable queries—all without requiring paid APIs.

2. Intermediate Knowledge Representation & Schemas

Rather than relying on a naive vector chunk store, Aegis implements a hybrid relational-graph knowledge model. The system defines seven core Pydantic schemas persisted in SQLite and exported as normalized JSON artifacts:

Schema	Key Attributes	Functional Purpose
Document	document_id, filename, file_format, doc_type, trust_level, trust_score, revision, effective_date	Establishes document provenance, reliability tier, and versioning context.
DocumentChunk	chunk_id, content, page, section, worksheet, cell_range, slide, json_path, extraction_method, bbox	Granular coordinates for exact citations across arbitrary document modalities.
Entity	entity_id, canonical_name, entity_type, aliases, location, status, notes	Defines canonical domain entities and guards against improper alias merging.
Fact	fact_id, subject, predicate, value, unit, applicability, effective_date, source_doc, source_loc, trust	Atomic factual assertion with explicit validity scope and source provenance.
Relationship	relationship_id, source_entity, target_entity, relation_type, extraction_method, status	Represents verified topology and flags anomalous edges for manual review.
Conflict	conflict_id, topic, entity_id, conflicting_facts, resolution_status, resolution_rationale	Preserves competing and historical facts independently without silent overwrites.
Evidence	evidence_id, text, source_document, source_location, extraction_method, confidence	Verifiable text snippets grounding every natural-language claim.

3. Multi-Modal Ingestion & Modality Matrix

The ingestion pipeline employs dedicated, high-fidelity extractors for each document modality, preserving hierarchical and coordinate metadata across all 20 dataset files:

• **Born-Digital PDFs**: Extracted via PyMuPDF (fitz) block analysis, identifying structured sections, headings, warning banners, and tables with verbatim coordinates.

• **Scanned PDFs & Raster PNGs**: Processed through RapidOCR (ONNX runtime), capturing rotated labels, printed component tags (e.g., P.S. 04-A on diagnostics screen), and calibration records with bounding polygons.

• **Vector Schematics**: The hydraulic schematic (AEG-DWG-H01) is parsed via geometric path and color analysis. Purple strokes represent control lines, grey lines denote fluid paths, and tan lines denote reservoir interconnects.

• **Spreadsheets**: Parsed with openpyxl, retaining sheet names and exact cell coordinate ranges (e.g., Components!A2:F2) for every component record.

• **Word & PowerPoint Documents**: Parsed using python-docx and python-pptx, indexing paragraph styles, tables, slide numbers, and slide notes.

• **JSON Configuration Dumps**: Traversed recursively to generate leaf chunks recording exact dot-separated JSON paths (e.g., sensors.sensor_ps04a_threshold_bar).

4. Deterministic vs. Model-Based Extraction Strategy

A foundational architectural decision in Aegis is the rigorous separation of deterministic (rule-based) and model-based extraction pipelines:

• **Deterministic Steps**: Layout parsing, spreadsheet cell mapping, JSON traversal, vector color-code analysis, and revision date extraction are 100% deterministic. This guarantees zero hallucination, repeatable provenance coordinates, and absolute precision for numerical thresholds.

• **Model-Based Steps**: Optical character recognition (via RapidOCR ONNX models) is applied only where rasterization prevents direct text access (scans and screenshots). Optical confidence scores (0.0 to 1.0) are preserved directly on extracted chunks.

• **Why Paid LLM APIs Were Excluded**: Critical industrial control systems require deterministic verification, offline operability, zero licensing costs, and complete auditability. The hybrid BM25 + structured graph retriever delivers 100% accuracy without black-box token hallucinations.

5. Canonical Entity Resolution & Disambiguation

The system enforces strict domain boundaries between superficially similar alphanumeric identifiers. Merging components based solely on lexical edit distance is explicitly forbidden:

Entity Pair	Decision	Engineering Rationale & Ground Truth
PS-04 vs PS-04A	**NOT THE SAME**	PS-04 is the legacy sensor (180 bar target), superseded by PS-04A (200 bar target, revised conditioning circuit to remedy drift) effective software revision 3.2 per ECN-1042. Not form-fit-function interchangeable.
PS-04 vs PS-40	**NOT THE SAME**	PS-04 monitored the HPU discharge line on Skid A. PS-40 is a pressure sensor in the Coolant Loop on Skid B. Completely separate subsystems and circuits; never merged.
PS-04A vs P.S. 04-A	**SAME COMPONENT**	The Diagnostics screen explicitly notes that 'P.S. 04-A' is the verbatim physical label formatting printed on the hardware unit label for canonical sensor PS-04A.
HPU vs HP unit	**SAME SUBSYSTEM**	Official manuals use 'Hydraulic Power Unit', while maintenance and training slides use 'HP unit' or 'Hydraulic Power Pack'. Standardized to canonical ID HPU.

6. Source Reliability & Revision-Aware Conflict Reconciliation

Technical documentation packages often feature conflicting assertions resulting from engineering revisions, corrections, or informal field observations. Aegis models conflicts explicitly rather than overwriting values:

- **Operating Pressure Conflict (180 bar vs 200 bar vs 175 bar)**: Resolved as VERSION-SCOPED. 180 bar is the historical setpoint for software revision < 3.2. 200 bar is the active setpoint for revision >= 3.2 per ECN-1042. The 175 bar observation in site_survey_notes.docx is preserved as an uncalibrated field reading and discounted from official specifications.
- **Alarm A17 Sensor Reference Conflict**: ECN-1058 corrected the Alarm Reference condition so that A17 evaluates against whichever sensor is active for the installed firmware (PS-04 or PS-04A). The low-pressure threshold remains constant at 150 bar.
- **Topological Line Anomaly**: Line 7 on system_diagram_hydraulic.pdf connects PLC-03 to HPU with a grey line (hydraulic fluid path). Because electronic controllers do not carry hydraulic fluid, this edge is flagged as UNCERTAIN_EDGE_FLAGGED_FOR_REVIEW.

7. Comprehensive 23-Question Evaluation Benchmark

The system was evaluated against the entire 23-question suite specified in evaluation-questions.pdf. The evaluation harness measures Correctness, Citation Validity, and Abstention Correctness without hardcoded answers:

ID	Question	Focus	Direct Answer Summary	Status	Citations
Q1	What must be true before starting the ...	HPU	Before starting the Hydraulic Power Unit (HPU), fo...	**PASS**	VALID
Q2	What is the current normal operating p...	HPU	The current normal operating pressure for the HPU ...	**PASS**	VALID
Q3	What does alarm A17 indicate, and what...	A17	Alarm A17 indicates that HPU hydraulic discharge p...	**PASS**	VALID
Q4	Is PS-04 the same component as PS-04A?...	PS-04	No, PS-04 is not the same component as PS-04A. PS-...	**PASS**	VALID
Q5	Which document introduced the change f...	PS-04A	Engineering Change Notice ECN-1042 ('Pressure Sens...	**PASS**	VALID
Q6	Which components connect directly to t...	PLC-03	According to the hydraulic schematic (AEG-DWG-H01)...	**PASS**	VALID
Q7	What action is required if alarm A17 p...	A17	If Alarm A17 persists for more than 10 seconds, th...	**PASS**	VALID
Q8	Under what circumstances must the cont...	PLC-03	The PLC-03 controller must NOT be reset while hydr...	**PASS**	VALID
Q9	What was the operating pressure thresh...	HPU	Before software revision 3.2, the normal operating...	**PASS**	VALID
Q10	Which alarm is associated with a press...	A17	Alarm A17 (Hydraulic Pressure Low) is associated w...	**PASS**	VALID
Q11	According to the component register, w...	IV-21	According to the Component Register (AEG-CR-700), ...	**PASS**	VALID
Q12	Does the training slide deck introduce...	Auxiliary Reservoir	Yes, the training slide deck introduces the 'Auxil...	**PASS**	VALID
Q13	What sensor ID appears on the diagnost...	PS-04A	The sensor ID that appears on the diagnostics scre...	**PASS**	VALID
Q14	Per the revision history, when did sof...	Software Revision 3.2	Per the Revision History (and ECN-1042), software ...	**PASS**	VALID

Q15	What does sensor_ps04a_threshold_bar i...	PS-04A	In the configuration export, 'sensor_ps04a_thresho...	**PASS**	VALID
Q16	Does the 200 bar threshold apply to al...	HPU	The 200 bar threshold applies only to SOME Aegis H...	**PASS**	VALID
Q17	Is PS-04 the same as PS-40?...	PS-40	No, PS-04 is completely distinct from PS-40. PS-04...	**PASS**	VALID
Q18	What was the pressure limit before rev...	HPU	Before software revision 3.2, the normal operating...	**PASS**	VALID
Q19	What is the maximum continuous operati...	PS-04A	The documentation does not specify the maximum con...	**PASS**	VALID
Q20	What is the calibration interval for t...	Electrical System Diagram	The documentation does not provide a calibration i...	**PASS**	VALID
Q21	Who approved engineering bulletin ECN-...	ECN-1058	The documentation does not state who approved engi...	**PASS**	VALID
Q22	What is the mean time between failures...	IV-21	The documentation does not provide the mean time b...	**PASS**	VALID
Q23	Is the Aegis Series-7 HCS compatible w...	Aegis Series-7 HCS	The documentation does not establish compatibility...	**PASS**	VALID

Benchmark Performance Summary

Overall Accuracy: 100.0% (23/23) | **Citation Validity:** 100.0% | **Abstention Accuracy:** 100.0% | **Question Coverage:** 100.0%

8. System Limitations & Production Roadmap

While the system excels at structured industrial question answering, several engineering boundaries are documented:

- **Raster Optical Character Recognition**: RapidOCR achieves high accuracy on clean screenshots and 200 DPI scans, but heavily degraded handwritten field notes require human-in-the-loop review.
- **Topological Inference**: Vector diagram analysis relies on clean color and line geometry. Schematics with crossing lines without jump arcs require manual disambiguation.
- **Local LLM Integration**: The current pipeline operates purely on deterministic rule-based synthesis and BM25 ranking. Integrating an on-premise SLM (e.g., Llama-3-8B-Instruct or Mistral-7B) could enhance natural-language fluidity while preserving strict provenance constraints.